

Regression Assumptions After Modeling

Executive summary

ISSUE / PROBLEM

The New York City Taxi & Limousine Commission engaged Automatidata to develop a model for predicting taxi fares. In this phase of the project, the Automatidata team produced the client's requested deliverable: a regression model.

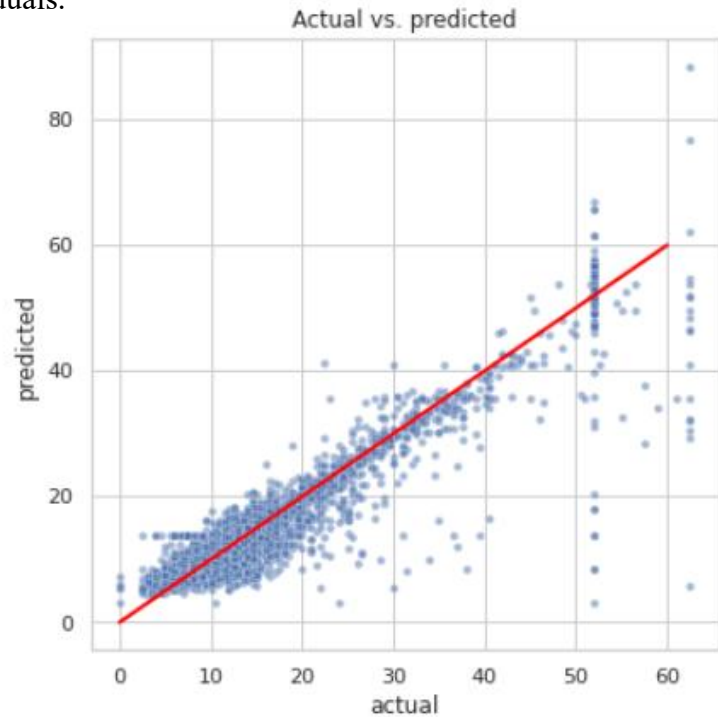
RESPONSE

The Automatidata team selected a multiple linear regression (MLR) approach based on the characteristics and distribution of the data. The MLR model successfully estimates taxi fares before a ride. Its strong performance on both training and test datasets indicates that the model is neither biased nor overfitted, with slightly better results observed on the test set.

IMPACT

Imputing outliers improved the model, particularly for the fare amount and trip duration variables. The linear regression model offers a reliable framework for predicting estimated taxi fares.

To demonstrate the effectiveness of the linear regression model, the Automatidata team included a scatter plot comparing predicted versus actual fare amounts. The model can reasonably predict taxi fares, and the accompanying notebook provides additional analysis of the model residuals.



The scatter plot shows a linear regression model plot illustrating predicted and actual fare amount for taxi cab rides.

Model metrics:

- Net model tuning resulted in:
 - ✓ R^2 0.87, meaning that 86.8% of the variance is described by the model.
 - ✓ MAE 2.1
 - ✓ MSE: 14.36
 - ✓ RMSE 3.8

KEY INSIGHTS

- Ride duration had the largest impact on fare amount, which aligns with expectations. The model indicated an average increase of \$7 per additional minute, though this figure may not be fully reliable due to high correlations among some features. Additional data from under-represented routes is recommended.

- The New York City Taxi and Limousine Commission can leverage these findings to develop an app that shows riders the estimated fare before their trip. The model delivers generally strong and reliable fare predictions, which can also support future modeling efforts.